Keshav Shivkumar

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EDUCATION

- **Master of Science (MS), Computer Science, 2024 | Rutgers University, New Brunswick, NJ**
  Specialization: *Machine Learning*
  CGPA: 3.83/4.00
- **Bachelor of Technology (B.Tech), Computer Science Engineering, 2022 | PES University, Bangalore, India**
  Specialization: *Machine Intelligence and Data Science*
  CGPA: 8.11/10.00

PROFESSIONAL EXPERIENCE

Teaching Assistant, Rutgers University, Piscataway, New Jersey **Jan 2023 – Present**

- TA for CS-170, Computer Applications for Business, which includes teaching students HTML, CSS, JavaScript, pseudocode, MS Excel, and databases.
- Set up office hours to help students, correct assignments and exams, and convey complex technical concepts in an accessible and engaging manner.

Summer Intern, Bloom Energy, Bangalore, Karnataka **Sept 2021 – Mar 2022**

- Developed a web application using Joomla CMS and Flask that streamlined the pipeline between Jenkins and the manufacturing services of Bloom Energy.
- Implemented a Jenkins Pipeline triggered by GitLab to run automated Selenium Java tests for the application, which sped up the process by 80%. Devised the logic and typed bash scripts to link Joomla and Jenkins.

TECHNICAL PROFICIENCIES

Programming Languages:	Python, Java, C, R
Web Development:	HTML, CSS, JavaScript, Django, ReactJS, Flask, Joomla, JSP
Other Relevant Skills:	SQL, Git, Linux, Data Analysis, PyTorch, OpenCV

PROJECTS

- **ViLT: Vision-and-Language Transformer** [Built with: Python (PyTorch)] {2023}: Open-source contribution to the ICML 2021 conference paper "ViLT" where the model is finetuned on a novel dataset, GQA.
- **Question Generation by Answering Questions** [Built with: Python (PyTorch)] {2023}: Unofficial implementation of the paper: Learning to Generate Question by Asking Question: A Primal-Dual Approach with Uncommon Word Generation.
- **Shopfinity** [Built with: SQL, JSP, JDBC, HTML, CSS, JavaScript] {2023}: A web application built to emulate the auctioning of vehicles.
- **Better, Smarter, Faster** [Built with: Python] {2022}**:** Improved on **Circle of Life** by building a Markov Decision Process and a regression neural network model from scratch to ensure ~100% win rate in the different graph environments.
- **Circle of Life** [Built with: Python] {2022}: Built probabilistic models of decision making for agents of different knowledge levels in a circular graph environment to catch a prey while also avoiding a predator.
- **Cloud-Based Skin Cancer Detection Application** [Built with: Python (TensorFlow, Keras), AWS and Android] {2021}: Developed a project with three phases: AWS (S3, SageMaker, EFS, Lambda, API Gateway), an Android application, and a DenseNet CNN model that classifies images of skin lesions. The result is returned as a JSON string in response to the API request sent by the user from the Android device.

PUBLICATIONS & RESEARCH

- K. Shivkumar, G. Swaminathan, T. Puvvadi, A. Shekar and S. Shilpa, "Cloud-Based Model for Detecting Skin Cancer," 2022 12th International Conference on Cloud Computing, Data Science & Engineering (Confluence), 2022, pp. 106- 111, doi: 10.1109/Confluence52989.2022.9734157.
- Identity theft - A Comprehensive analysis [2022]: A research survey comparing different approaches and methods utilized to prevent and respond to identity theft effectively.